

54 | OVERVIEW



Beyond Renewable Electricity: Comprehensive Climate Strategies for Remote Economies

Abstract

In 2023, New Zealand's government partnered with investment manager BlackRock on a NZ\$2 billion renewable energy fund. Climate scientist Kevin Trenberth argued that private investment in renewable electricity is essential but not enough. As an isolated country that depends on shipping, flights, exports, and tourism, New Zealand needs a comprehensive strategy. He proposed ten areas to address.

Introduction

New Zealand already generates over 80% of its electricity from renewable sources, mainly hydropower plus wind and solar, and aims for 100% by 2035. That sounds like a climate success story. But a remote island nation faces special challenges. Its economy relies on exports and tourism, which depend on ships and planes still powered by fossil fuels. Carbon pricing and carbon-based trade tariffs could raise the cost of both imports and exports. Storms and droughts can disrupt food production. Trenberth argued that meeting the long-term climate threat will require a comprehensive, collaborative approach that looks well beyond the electricity grid.

How Do Scientists Study This?



FROM THE ARTICLE

New Zealand is isolated and relies on shipping and air travel, making it vulnerable to carbon pricing. Map: Wikimedia, CC BY-SA.

This lesson analyzes a policy proposal. Trenberth examined New Zealand's emissions sources and vulnerabilities, including transport, which produces 17% of gross emissions, and global shipping, which, if it were a country, would rank sixth in total emissions. He considered how intermittent wind and solar power can be paired with hydropower dams acting as batteries. He reviewed options for carbon offsets, forestry, biofuels such as torrefied wood, and land use. He then organized his recommendations into ten broad areas, weighing each for its ability to cut emissions, build resilience, and strengthen the economy.

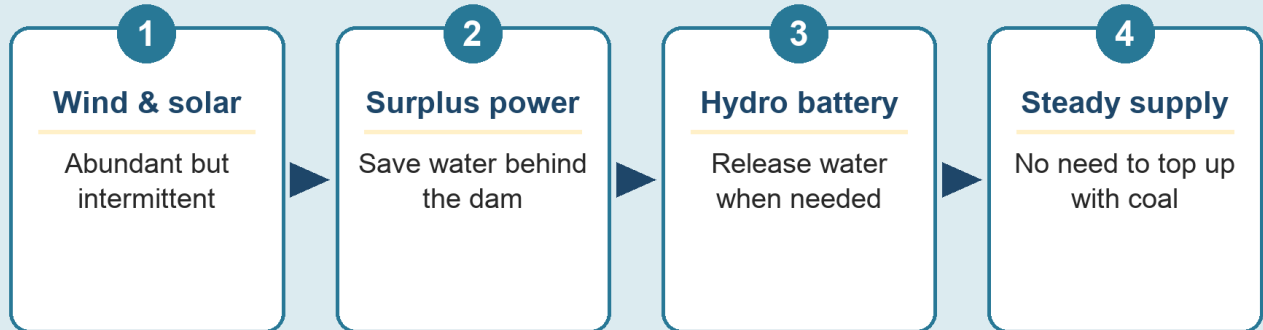


FROM THE ARTICLE

Because wind and solar power are intermittent, they must be integrated with hydropower. Photo: Shutterstock/Dmitry Pichugin.

Explore the Evidence

Pairing renewables with hydropower



Identify vulnerabilities

Examine dependence on shipping, flights, tourism, and exports.

Find emission sources

Transport, industry heat, and fossil-fueled travel remain big contributors.

Propose broad solutions

Ten areas span energy, travel, forests, cities, fuels, and land use.

What Did They Discover?

Trenberth's ten areas are: maximize renewable energy through a publicly owned entity that uses hydro as a battery; rethink travel, including avoiding unnecessary trips; reduce shipping emissions, following the International Maritime Organization's goal of net zero close to 2050; consider trains versus planes; prepare for tourism declines; improve carbon offsets, recognizing trees are temporary; practice strategic forestry with native trees; green cities to counter urban heat islands; develop biofuels like torrefied wood, which already replaces coal at the Huntly power station but is imported; and create incentives for regenerative farming and agroforestry. Overall, he argued, New Zealand must become more self-sufficient, processing and manufacturing more at home.

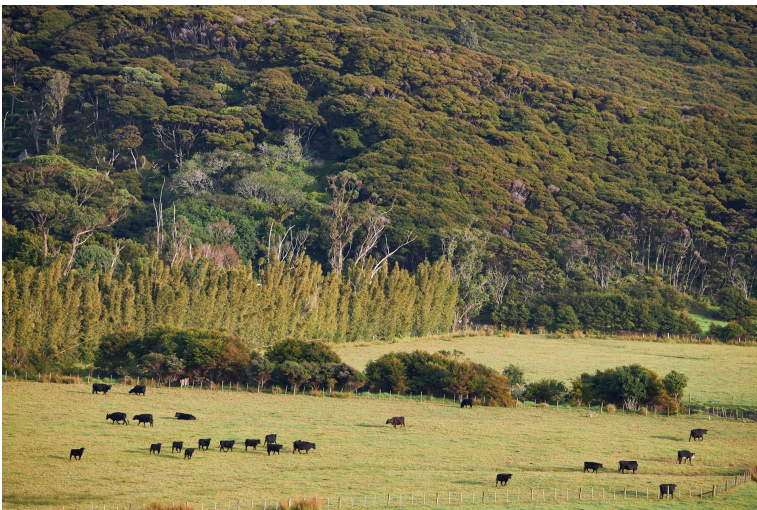
New Zealand's climate challenge

80%+ of electricity already from renewable sources

17% of gross emissions from transport

6th shipping's rank if it were a country's emissions

Indigenous trees grow slowly but can lock up carbon for more than a century, as long as forests are protected from fire and disease.



FROM THE ARTICLE

Planting native trees on farms helps store carbon, improves soil health, and provides extra income. Photo: Getty Images/Mark Meredith.

Takeaways

Electricity isn't everything

Transport, shipping, industry, and tourism also need climate solutions.

Isolation adds risk

Remote economies depend on fossil-fueled ships and planes and face carbon tariffs.

Think in systems

Energy, forests, cities, farms, and trade must be planned together.

Key Terms

Decarbonization

Reducing carbon dioxide emissions, especially by replacing fossil fuels with clean energy.

Intermittent energy

Power sources, like wind and solar, that vary with weather and time of day.

Pumped storage hydropower

Storing energy by pumping water uphill behind a dam and releasing it to generate electricity when needed.

Torrefaction

Heating biomass without oxygen to create a dense, dry fuel that can replace coal.

Agroforestry

Integrating trees with crops or livestock to store carbon, improve soil, and diversify income.

Activity

Island Nation Climate Plan

Students act as advisers to an island nation and prioritize Trenberth's ten strategies.

Materials: Cards for each of the ten strategies; Criteria sheet: emissions cut, cost, jobs, resilience; Poster paper

1. Read each strategy card and rate it from 1 to 5 on each criterion.
2. Choose the top three strategies for the first five years and explain why.
3. Identify one strategy that depends on international action, such as shipping rules.
4. Explain how your plan reduces dependence on imported fossil fuels.
5. Present your plan and compare priorities with other groups.

References

Trenberth, K. (2023). Meeting the long-term climate threat takes more than private investment: 10 ways NZ can be smart and strategic. *The Conversation*. <https://doi.org/10.64628/AA.tgvncu5pe>